

# INTERNSHIP REPORT

## 1. Introduction

This internship provided me with an opportunity to gain practical experience in Data Analytics using Microsoft Power BI. The primary objective of the internship was to analyze a Twitter dataset and create meaningful visualizations that could help understand user engagement, interaction patterns, and tweet performance.

Throughout the internship, I worked with real-world social media dataset containing information such as Tweet ,id , time ,impressions ,engagements ,engagement rate ,retweets ,replies ,likes ,user profile clicks ,url clicks ,hashtag clicks ,detail expands ,permalink clicks ,app opens ,app installs ,follows ,email tweet ,dial phone ,media views and media engagements .Using Power BI, I transformed raw data into interactive dashboards and reports that provided valuable insights into user behavior and tweet performance.

This internship allowed me to apply theoretical concepts of data analytics in a practical environment while improving my technical and analytical skills.

## 2. Background

Social media platforms generate a huge volume of data every day. Organizations use this data to understand customer behavior, measure engagement, evaluate content performance, and make informed decisions.

For this project, a Twitter dataset was provided that contained multiple metrics related to tweet interactions and engagement. The goal was to analyze the data using Power BI and create dashboards based on specific requirements and filtering conditions.

The project required the implementation of various analytical techniques including:

- Data cleaning and transformation
- Creation of calculated columns
- DAX measures
- Time-based visual controls
- Conditional filtering

- Dashboard design and formatting
- Trend and performance analysis

Each task focused on a different aspect of Twitter analytics and required the implementation of specific conditions to ensure accurate analysis.

### **3. Learning Objectives**

The key learning objectives of this internship were:

- To understand the fundamentals of data analytics.
- To learn how to use Microsoft Power BI for dashboard development.
- To gain hands-on experience in creating visual reports from raw datasets.
- To understand ETL(Extract-Transform-Load) process.
- To understand data cleaning and transformation techniques.
- To learn how to create calculated columns and DAX measures.
- To analyze social media engagement metrics and identify trends.
- To improve problem-solving and analytical thinking abilities.
- To develop professional reporting and dashboard presentation skills.

### **4. Activities and Tasks Performed**

During the internship, I completed six analytical tasks using the provided Twitter dataset.

#### **Task 1: Tweet Interaction Breakdown by Category**

A clustered bar chart was created to analyze user interactions across different tweet categories such as Media Tweets, Link Tweets, and Hashtag Tweets.

The chart displayed:

- URL Clicks
- User Profile Clicks
- Hashtag Clicks

Several conditions were applied:

- Visualization visible only between 3 PM IST and 5 PM IST.
- Tweet Date must be an even number.
- Tweet Word Count must be greater than 40.
- Tweets must contain at least one interaction type.

### **Observation**

After applying the mandatory condition of Tweet Word Count greater than 40, no records in the dataset satisfied the requirement. As a result, the visualization became blank.

This was not an implementation error. The filter was intentionally retained because the task instructions required all conditions to be applied exactly as specified. The observation highlights a limitation in the dataset where no tweets met the required word count condition.

### **Task 2: Engagement Rate Comparison**

A comparison chart was developed to analyze the engagement rate of tweets that included app opens versus tweets without app opens.

Conditions applied:

- Tweets posted between 9 AM and 5 PM.
- Weekday tweets only.
- Tweet impressions must be even numbers.
- Tweet dates must be odd numbers.
- Character count greater than 30.
- Tweets containing the letter "D" were excluded.

- Visualization visible only during 12PM IST TO 6PM IST and 7AM IST TO 11AM IST.

### **Observation:**

After applying all the required filters, only tweets from the "No App Open" category were available in the dataset. No tweets from the "App Open" category met all the specified conditions. Therefore, the chart displayed only the "No App Open" category.

The average engagement rate after applying all filters was 0.02135. This shows the engagement level of the tweets that satisfied the given conditions.

This was not an error in the dashboard. The result occurred because the dataset did not contain any "App Open" tweets that matched all the required filters.

The analysis demonstrated how engagement metrics vary after applying multiple business conditions. The task also highlighted the importance of validating filtered datasets before drawing conclusions from visualizations.

### **Task 3: Media Interaction by Day of Week**

A dual-axis chart was created to compare Media Views and Media Engagements across different days of the week.

Additional conditions included:

- Last quarter data only(Q4).
- Even tweet impressions.
- Odd tweet dates.
- Character count above 30.
- Tweets containing the letter "H" were excluded.
- Visibility controlled through 3PM IST TO 5PM IST and 7AM IST TO 11AM IST.

### **Observation**

After applying all the required filters, it was observed that Thursday had the highest media interaction, with 120 Media Views and 104 Media Engagements. This indicates that users were most active in viewing and engaging with media content on that day.

It was also noticed that Tuesday and Sunday were not displayed in the chart because no records from those days met all the specified filtering conditions.

Overall, the chart showed that days with higher media views generally had higher media engagements, suggesting that users who viewed media content were also more likely to interact with it.

The visualization helped identify days with higher media interactions and provided insights into audience engagement patterns. The comparison between media views and media engagements showed how content consumption influences user interaction.

#### **Task 4: Replies, Retweets, and Likes Comparison**

A bar chart was developed to compare the total number of:

- Replies
- Retweets
- Likes

for tweets posted between June and August 2020.

Simple SUM aggregation was used for all metrics.

#### **Observation**

This visualization compared the SUM of replies, retweets, and likes for tweets posted between June and August 2020. It was observed that likes contributed the highest level of engagement, followed by retweets and replies.

Among the three months, July showed the highest engagement levels, with the highest number of likes, retweets, and replies compared to June and August. This suggests that tweets posted during July generated stronger audience interaction and performed better overall.

In contrast, June and August recorded lower engagement levels, indicating comparatively reduced audience activity during those months.

The analysis provided a clear understanding of overall engagement distribution and highlighted which engagement metric contributed most significantly during the selected period.

### **Task 5: Monthly Engagement Rate Trend**

A line chart was created to analyze the monthly trend of average engagement rates.

The chart compared:

- Tweets with Media Content
- Tweets without Media Content

Month was used as the time dimension and average engagement rate as the primary metric.

#### **Observation**

This visualization compared the average engagement rate of tweets with media content and tweets without media content across different months.

It was observed that for tweets without media content, the highest engagement rate was recorded in September, while the lowest engagement rate was observed in June. In contrast, tweets with media content achieved their highest engagement rate in June and the lowest in September

The trend analysis showed differences in engagement patterns between media and non-media content tweets. The visualization also helped identify periods of increased or decreased audience engagement throughout the Months.

### **Task 6: Top 10 Tweets by Engagement**

A chart was developed to identify the Top 10 Tweets based on total engagement calculated using likes and retweets.

The following conditions were applied:

- Weekend tweets excluded.
- Tweet impressions must be even.
- Tweet dates must be odd.

- Tweet word count less than 30.
- Visualization visible only between 3 PM IST and 5 PM IST.
- User profile information displayed alongside tweet performance.

## **Observation**

This visualization identified the Top 10 tweets based on Total Engagement, which was calculated using the sum of likes and retweets. The analysis included only tweets that satisfied all the required conditions.

The chart highlighted the tweets that generated the highest audience interaction and helped identify the most successful content in the dataset.

Along with the tweet details, User ID and User Profile Clicks were displayed to provide additional information about the users associated with the top-performing tweets. Higher profile click counts indicated that users were interested in learning more about the tweet creators after engaging with their content.

This task helped identify the highest-performing tweets and understand the characteristics of successful content. The visualization provided insights into which tweets generated the greatest audience response.

## **5. DAX Measures and Calculated Columns Implemented**

- Chart Visibility Measure (Time-Based)
- Total Engagement Measure
- Tweet Category Classification
- Tweet Word Count Calculation
- Character Count Calculation
- Month Sort Order
- Weekday Sort Order
- Quarter Identification

- Media Content Classification
- Tweet Impression (Even/Odd)
- Tweet Date (Even/Odd)
- Tweet Time Filter
- Excluding Tweet with letter 'H'
- Excluding Tweet with letter 'D'

## **6. Skills and Competencies Developed**

During the internship, I gained both technical and analytical skills, including:

### **Technical Skills**

- Microsoft Power BI
- Data Visualization
- Dashboard Design
- Data Transformation
- DAX Measures
- Calculated Columns
- Filtering and Slicers
- Time-Based Visual Controls

### **Analytical Skills**

- Data Interpretation
- Trend Analysis
- Performance Measurement
- Problem Solving
- Critical Thinking
- Decision Making Based on Data



## **Professional Skills**

- Report Writing
- Documentation
- Attention to Detail
- Project Management
- Communication of Insights

## **7. Key Highlights of the Project**

- Successfully developed six Power BI dashboards based on Twitter analytics.
- Implemented complex filtering conditions according to business requirements.
- Applied time-based visibility controls using DAX measures.
- Created custom sorting for months and weekdays.
- Developed calculated columns for tweet categorization and analysis.
- Validated all dashboard outputs against task requirements.
- Identified and documented dataset anomalies instead of modifying business rules.
- Improved dashboard formatting and visualization design to enhance readability and user experience.

## **8. Feedback and Evidence**

The dashboards developed during the internship successfully met the requirements provided for each task. Screenshots of completed dashboards and the Power BI project file serve as evidence of the work completed.

The project demonstrated the ability to apply business rules, perform data analysis, create interactive visualizations, and communicate findings effectively through reports and dashboards.

## **9. Challenges Faced and Solutions Implemented**

Several challenges were encountered during the project.

### **Challenge 1: Time-Based Visualization Control**

Implementing charts that appeared only during specific time intervals required creating DAX measures and applying visual-level filters.

#### **Solution:**

Custom DAX measures were developed to control chart visibility based on current system time.

### **Challenge 2: Sorting Months and Days Correctly**

Power BI initially sorted month names and weekdays alphabetically.

#### **Solution:**

Custom sort columns were created and applied to ensure proper ordering.

### **Challenge 3: Applying Multiple Conditions Simultaneously**

Many tasks required multiple filters to be applied together.

#### **Solution:**

Each filter condition was carefully implemented and validated individually before combining them.

### **Challenge 4: Empty Visualization After Filtering**

In Task 1, the visualization became blank after applying the required Word Count greater than 40

#### **Solution:**

The filter was retained because it was a required business condition. The anomaly was documented in the report rather than modifying the requirement.

## **10. Outcomes and Impact**

This internship significantly improved my understanding of data analytics and business intelligence concepts. I learned how to transform raw datasets into meaningful visualizations and how to communicate insights through dashboards.

The project strengthened my ability to:

- Work with real-world datasets.
- Apply business rules through calculations and filters.
- Create professional dashboards.
- Analyze trends and engagement patterns.
- Present findings in a structured and meaningful way.

The experience also strengthened my confidence in handling analytical projects and solving practical business problems using data. It also improved my ability to interpret business requirements and translate them into meaningful visualizations using Power BI.

## **11. Conclusion**

The internship provided valuable hands-on experience in Data Analytics and Business Intelligence using Microsoft Power BI. Through the successful completion of six Twitter analytics dashboards, I gained practical knowledge of data visualization, DAX calculations, filtering techniques, dashboard design, and analytical reporting.

One of the most important lessons learned during the project was the importance of implementing business requirements exactly as specified, even when the resulting visualization contains no data. This reinforced the significance of data validation and accurate documentation during analytical projects.

One of the most valuable aspects of this internship was gaining hands-on experience with real-world data. It helped me understand that successful data analysis involves not only creating visualizations but also validating data, handling anomalies, and accurately documenting observations.

Overall, the internship successfully bridged the gap between theoretical knowledge and practical application while helping me develop technical, analytical, and professional skills that will support my future career in Data Analytics and Business Intelligence.